Ganesh Chandrasekar

AI/ML Engineer | Machine Learning | NLP | Generative AI | Software Engineering

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PROFESSIONAL SUMMARY

AI/ML Engineer with a Master of Computer Science and experience across machine learning, NLP, retrieval, and software development. Work includes model training and evaluation, retrieval and LLM systems, APIs, and user-facing applications across both research and product environments.

PROFESSIONAL EXPERIENCE

Research Assistant Sep 2023 – Aug 2026
CLaC Lab, Concordia University *Montreal, Canada*

- Built biomedical QA pipelines that retrieved and reranked evidence before generation, then validated whether the final answer was supported by the cited sources.
- Separated evaluation by retrieval, answer quality, grounding, and citation behavior to identify which stage was responsible when a system failed.
- Trained and compared BERT-family, CRF, hierarchical, and PPO models for NLP tasks, with attention to class imbalance, per-label performance, and failure cases.
- Ran quantized language-model inference with **vLLM** and Slurm on V100/A100 GPUs for repeatable large-scale evaluation experiments.

Solution Developer Jul 2022 – Aug 2023
OrangeScape Technologies Pvt. Ltd. *Chennai, India*

- Worked across upstream and downstream procurement workflows in Kissflow Procurement Cloud, including purchasing, supplier management, approvals, and source-to-pay processes.
- Built and integrated GCP-backed services, connecting frontend workflows with backend APIs, validations, business rules, and external/internal service integrations.
- Prototyped proof-of-concepts and automation ideas for product workflows, then helped take selected changes through implementation, testing, and release.
- Supported product testing, troubleshooting, and production management by working with backend, product, and QA teams to resolve workflow and integration issues.

Software Engineering Intern May 2022 – Jul 2022
Basik Marketing Pvt. Ltd. *Chennai, India*

- Built a Python and Node.js broadcast overlay that consumed live match state and updated scores, event counters, and on-screen elements during esports broadcasts.

Undergraduate Research Assistant Jan 2021 – Jan 2022
Solarillion Foundation *Chennai, India*

- Built a two-stage flight-delay prediction pipeline from historical flight data, preparing features, training supervised models, and comparing performance across the pipeline.
- Designed citation-intent classification experiments from text preprocessing and feature extraction through model training, label-level evaluation, and error analysis.

Software Engineering Intern Jan 2021 – Jun 2021
PALS: An Initiative by Alumni Fraternity of IITs *India*

- Built an interactive virtual-lab simulator for the Synchro Transmitter/Receiver experiment, reproducing the experiment flow and expected outputs in software.

Ganesh Chandrasekar | **AI/ML Engineer**

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, TypeScript/JavaScript, Java

AI & Machine Learning: PyTorch, scikit-learn, Hugging Face Transformers, BERT, BioBERT, CRF, NLP, Text Classification, NER

Retrieval & AI Systems: BM25, Pyserini/Lucene, MedCPT, FAISS, Cross-Encoder Reranking, RAG, LangChain, LangGraph, MCP, smolagents, Qwen

Software & Cloud: REST APIs, Flask, Node.js, Next.js, Docker, GCP, AWS, Linux, Slurm, Git, vLLM, Ollama, Supabase

PROJECTS

BioEvidence AI: Evidence-Grounded Biomedical Question Answering *Live Demo*↗

Python | Pyserini/Lucene | RAG | Local LLMs | Next.js | Supabase | REST APIs

- Turned the biomedical QA pipeline into a working application with separate retrieval, reranking, generation, and citation-validation stages that could be inspected independently.
- Connected the Python backend to PubMed/NCBI, Europe PMC, and ClinicalTrials.gov APIs, then added Supabase authentication/application data and a Next.js interface.

Agentic and Non-Agentic Multi-Hop Medical QA *GitHub*↗

Python | smolagents | LLM Agents | Retrieval | Tool Use

- Implemented an agentic retrieval loop where the model could search for evidence, review what it had found, and decide whether to continue or answer, with a 15-step execution cap.
- Compared it with controlled Wikipedia and Wikipedia + PubMed retrieval pipelines to study how iterative tool use and retrieval strategy changed answer quality.

Biomedical Negation Detection *GitHub*↗

Python | PyTorch | BERT / BioBERT / ClinicalBERT | CRF | PPO

- Fine-tuned BERT, BioBERT, and ClinicalBERT for cue and scope tagging, then compared CRF and hierarchical variants under severe class imbalance.
- Trained a PPO-based sequence-labeling policy from annotated spans and compared it with supervised approaches using aggregate and label-level evaluation to understand minority-class behavior.

EDUCATION

Master of Computer Science (Thesis) Sep 2023 – Aug 2026
Concordia University *Montreal, Canada*
Thesis: *Evidence-Grounded Biomedical Question Answering: Design, Analysis, and Evaluation*

Bachelor of Engineering in Computer Science and Engineering Jun 2018 – Apr 2022
KCG College of Technology, Anna University *Chennai, India*
CGPA: 8.81/10, First Class with Distinction

PUBLICATIONS

TREC 2025 BioGen | *Modular Sparse and Dense Retrieval for Evidence-Constrained Biomedical Question Answering*

BioCreative IX, IJCAI 2025 | *Agentic and Non-Agentic Multi-Hop Systems for Medical Question Answering*